AI-Assisted Systematic Optimization of the LAMMPS Molecular Dynamics Simulator via Large Language Model Coding Agents

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Supplementary Materials

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Contents

Supplementary Table S1: Complete List of 35 New OMP Pair Variants

All styles below were generated by Claude Code, compiled with zero new MSVC warnings, and verified bit-identical to their serial counterparts by a 50-step NVE thermo comparison (`thermo_modify format float %20.15g`). Files are located in `src/OPENMP/` with the naming convention `pair_[style_name]_omp.{h,cpp}`.

#	Package	Standard style	OMP variant	Loop type	New register
1	EXTRA-PAIR	lj/cut/soft	lj/cut/soft/omp	half-Newton	—
2	EXTRA-PAIR	lj/cut/coul/long/soft	lj/cut/coul/long/soft/omp	half-Newton	—
3	EXTRA-PAIR	lj/class2/soft	lj/class2/soft/omp	half-Newton	—
4	EXTRA-PAIR	lj/cut/coul/cut/soft	lj/cut/coul/cut/soft/omp	half-Newton	—
5	EXTRA-PAIR	lj/cut/coul/dsf/soft	lj/cut/coul/dsf/soft/omp	half-Newton	—
6	EXTRA-PAIR	morse/soft	morse/soft/omp	half-Newton	—
7	EXTRA-PAIR	coul/cut/soft/gapsys	coul/cut/soft/gapsys/omp	half-Newton	—
8	EXTRA-PAIR	nm/cut/split	nm/cut/split/omp	half-Newton	—
9	EXTRA-PAIR	born/coul/dsf	born/coul/dsf/omp	half-Newton	—
10	EXTRA-PAIR	born/coul/wolf/cs	born/coul/wolf/cs/omp	half-Newton	—
11	EXTRA-PAIR	bpm/spring	bpm/spring/omp	half-Newton	—
12	EXTRA-PAIR	dpd/fdt	dpd/fdt/omp	half-Newton	random seed/thr
13	EXTRA-PAIR	dpd/fdt/energy	dpd/fdt/energy/omp	half-Newton	random seed/thr
14	EXTRA-PAIR	lj/relres/soft	lj/relres/soft/omp	half-Newton	—
15	EXTRA-PAIR	coul/diel	coul/diel/omp	half-Newton	—
16	EXTRA-PAIR	momb	momb/omp	half-Newton	—
17	EXTRA-PAIR	coul/streitz	coul/streitz/omp	half-Newton	charge_thr buf
18	EXTRA-PAIR	extep	extep/omp	half-Newton	—
19	DIELECTRIC	lj/cut/coul/cut/dielectric	lj/cut/coul/cut/dielectric/omp	full (no Newton)	ef_tally_thr
20	DIELECTRIC	lj/cut/coul/long/dielectric	lj/cut/coul/long/dielectric/omp	full (no Newton)	ef_tally_thr
21	DIELECTRIC	lj/cut/coul/msm/dielectric	lj/cut/coul/msm/dielectric/omp	full (no Newton)	ef_tally_thr
22	DIELECTRIC	coul/long/dielectric	coul/long/dielectric/omp	full (no Newton)	ef_tally_thr
23	DIELECTRIC	lj/long/coul/long/dielectric	lj/long/coul/long/dielectric/omp	full (no Newton)	ef_tally_thr
24	DIELECTRIC	lj/cut/coul/debye/dielectric	lj/cut/coul/debye/dielectric/omp	full (no Newton)	ef_tally_thr
25	DIELECTRIC	coul/cut/dielectric	coul/cut/dielectric/omp	full (no Newton)	ef_tally_thr
26	KSPACE	lj/long/coul/long	lj/long/coul/long/omp	half-Newton	—
27	KSPACE	buck/long/coul/long	buck/long/coul/long/omp	half-Newton	—
28	KSPACE	lj/long/tip4p/long	lj/long/tip4p/long/omp	half-Newton	tip4p_h_thr
29	KSPACE	lj/cut/tip4p/long	lj/cut/tip4p/long/omp	half-Newton	tip4p_h_thr
30	KSPACE	lj/long/dipole/long	lj/long/dipole/long/omp	half-Newton	—

(continued on next page)

(Table S1 continued...)

#	Package	Standard style	OMP variant	Loop type	New register
31	EXTRA-PAIR	buck/coul/cut/soft	buck/coul/cut/soft/omp	half-Newton	—
32	EXTRA-PAIR	buck/coul/long/soft	buck/coul/long/soft/omp	half-Newton	—
33	EXTRA-PAIR	lj/charmm/coul/long/soft	lj/charmm/coul/long/soft/omp	half-Newton	—
34	EXTRA-PAIR	lj/charmm/coul/charmm/soft	lj/charmm/coul/charmm/soft/omp	half-Newton	—
35	EXTRA-PAIR	coul/ctip	coul/ctip/omp	half-Newton	charge_iter_thr

Notes: - *Loop type "full"*: DIELECTRIC styles use a full neighbor list (all neighbors i→j, not half-list) because the dielectric boundary condition requires summing contributions from all neighbors of i. No Newton's third law write-back is applied. - *ef_tally_thr*: The `ef_tally_thr()` function (added to `src/OPENMP/thr_omp.h`) accumulates per-thread electric field contributions `efield[i][k]` in addition to energy/virial. - *tip4p_h_thr*: For TIP4P water, the hydrogen force redistribution (from the virtual M-site) is thread-safe only with per-thread hydrogen force buffers. - *random seed/thr*: DPD stochastic forces use per-thread random number generators seeded with `seed + thread_id` to ensure deterministic but uncorrelated thread-local sequences.

Supplementary Table S2: A-3 Modification Log (Representative Subset)

Full log contains 102 entries. Representative entries spanning each category:

File	Category	Modification	Bit-id.	Pair improv.
pair_lj_cut.cpp	Simple LJ	`_noalias` + `fxtmp/fytmp/fztmp`		+4.2% (4k FCC)
pair_lj_expand.cpp	Simple LJ	same		+3.8%
pair_coul_cut.cpp	Simple Coul	same		+4.1%
pair_coul_wolf.cpp	Comp. Coul	same		+5.2%
pair_born_coul_dsf.cpp	Comp. Coul	same		+5.7%
pair_born_coul_wolf_cs.cpp	Comp. Coul	same		+4.9%
pair_lj_long_coul_long.cpp	Long-range	same (real-space loop only)		+3.5%
pair_lj_cut_coul_long_dielectric.cpp	DIELECTRIC	same + `ef_tmp` for field accum		+4.0%
pair_momb.cpp	Specialty	same		+7.5%
pair_coul_ctip.cpp	Charge-iter	same (inner pair loop only)		+5.7%
pair_dispersion_d3.cpp	Multi-pass	NOT applicable — multi-pass loop structure	n/a	—
pair_eam.cpp	Multi-body EAM	NOT applicable — loop-carried ϕ_i accumulation	n/a	—
pair_tersoff.cpp	Multi-body	NOT applicable — three-body sums	n/a	—
pair_snap.cpp	ML potential	NOT applicable — internal vectorization	n/a	—

Note on non-applicable styles: Multi-body potentials (EAM, Tersoff, SW, ReaxFF) use multi-pass loops where force contributions from one atom depend on previously visited neighbors; the `fxtmp` pattern requires a single-pass inner loop. ML potentials (SNAP, ACE, DeePMD) implement their own BLAS-based or SIMD-specialized inner operations that supersede simple pointer qualification.

Supplementary Table S3: OPLS-AA Force Field Parameters for Battery Electrolyte Solvents

All parameters follow Jorgensen *et al.* (1996) OPLS-AA conventions. LJ potential: U_LJ = $4\varepsilon[(\sigma/r)^{12}-(\sigma/r)^{6}]$; harmonic bond: U_bond = $K(r-r_0)^2$; harmonic angle: U_angle = $K(- {}_0)^2$; OPLS dihedral: U_torsion = $½V_1(1+\cos\ ) + ½V_2(1-\cos 2\ ) + ½V_3(1+\cos 3\ ) + ½V_4(1-\cos 4\ )$.

Atom Types (shared across EC and PC)

Atom type	Description	σ (Å)	ε (kcal/mol)	Partial charge q (e)
C_co3	Carbonyl C in carbonate	3.750	0.1050	+0.76
O_co3	Carbonyl O (C=O)	2.960	0.2100	−0.50
O_ete	Ether O (ring C−O−C)	3.000	0.1700	−0.35
C_sp3	Aliphatic CH_2 in ring	3.500	0.0660	+0.10
H_c	H on aliphatic C	2.500	0.0300	+0.06

EC molecule charge neutrality: C_co3 + 2×O_co3 + 2×O_ete + 2×C_sp3 + 4×H_c = +0.76 + 2×(−0.50) + 2×(−0.35) + 2×(+0.10) + 4×(+0.06) = +0.76 − 1.00 − 0.70 + 0.20 + 0.24 = 0.00

Atom Types (DME: 1,2-dimethoxyethane)

Atom type	Description	σ (Å)	ε (kcal/mol)	Partial charge q (e)
C_sp3(CH_3)	Terminal methyl C	3.500	0.0660	+0.022
O_ete	Ether O	3.000	0.1700	−0.400
C_sp3(CH_2)	Central methylene C	3.500	0.0660	+0.164
H_c(CH_3)	H on methyl	2.500	0.0300	+0.034
H_c(CH_2)	H on methylene	2.500	0.0300	+0.056

DME molecule charge neutrality: 2×C_sp3(CH_3) + 2×O_ete + 2×C_sp3(CH_2) + 6×H_c(CH_3) + 4×H_c(CH_2) = 2×(+0.022) + 2×(−0.400) + 2×(+0.164) + 6×(+0.034) + 4×(+0.056) = 0.044 − 0.800 + 0.328 + 0.204 + 0.224 = 0.000

Bond Parameters

Bond	K (kcal/mol/Å^2)	r_0 (Å)
C_co3−O_co3	570.0	1.229
C_co3−O_ete	415.0	1.344
C_sp3−O_ete	320.0	1.430
C_sp3−C_sp3	224.0	1.529
C_sp3−H_c	340.0	1.090

Angle Parameters

Angle	K (kcal/mol/rad^2)	$_0$ (°)
O_co3–C_co3–O_ete	90.0	124.4
C_co3–O_ete–C_sp3	70.0	109.5
O_ete–C_sp3–C_sp3	70.0	111.5
O_ete–C_sp3–H_c	55.0	109.5
H_c–C_sp3–H_c	35.0	107.8
C_sp3–C_sp3–H_c	37.5	110.7

Dihedral Parameters (OPLS style, V_1–V_4 in kcal/mol)

Dihedral	V_1	V_2	V_3	V_4
O_co3–C_co3–O_ete–C_sp3	1.950	2.250	0.000	0.000
C_co3–O_ete–C_sp3–C_sp3	0.650	−0.250	0.670	0.000
C_co3–O_ete–C_sp3–H_c	0.000	0.000	0.760	0.000
O_ete–C_sp3–C_sp3–O_ete	−0.550	−0.550	0.000	0.000
O_ete–C_sp3–C_sp3–H_c	0.000	0.000	0.468	0.000
H_c–C_sp3–C_sp3–H_c	0.000	0.000	0.318	0.000

Supplementary Table S4: Li^+ Parameters and Lorentz–Berthelot Mixed Pair Coefficients

Li^+ source: Joung & Cheatham (2008), SPC/E water parameterization.
Mixing rules: $\sigma_{12} = (\sigma_1 + \sigma_2)/2$ (arithmetic); $\varepsilon_{12} = \sqrt{(\varepsilon_1 \times \varepsilon_2)}$ (geometric).

Parameter	Li^+	Reference
σ (Å)	1.4094	Joung & Cheatham 2008
ε (kcal/mol)	0.3367	Joung & Cheatham 2008
q (e)	+1.0	formal charge

Lorentz–Berthelot Mixed Coefficients: Li^+ × OPLS-AA Solvent Atoms

Pair	σ_mix (Å)	ε_mix (kcal/mol)	Context
Li–Li	1.4094	0.33670	self-pair (irrelevant, single Li+)
Li–C_co3	2.5797	0.18802	EC/PC carbonyl carbon
Li–O_co3	2.1847	0.26591	EC/PC carbonyl oxygen
Li–O_ete	2.2047	0.23921	EC/PC/DME ether oxygen
Li–C_sp3	2.4547	0.14898	EC/PC/DME aliphatic carbon
Li–H_c	1.9547	0.10049	all H atoms

Note on Coulombic interactions: The `lj/cut/coul/long/soft` potential scales *both* the LJ and Coulomb interactions by λ simultaneously. The Li^+ charge (+1.0 e) interacts with OPLS-AA partial charges via PPPM long-range Coulomb. The dominant contribution to $dU/d\lambda$ is the

Coulombic Li^+–O_co3 and Li^+–O_ete attraction (solvation coordination shell). A +1 net charge in the simulation box is handled by PPPM with a uniform background charge correction, which introduces a small but systematic correction to the absolute ΔG (estimated < 0.1 kcal/mol for the box sizes used) without affecting relative comparisons between solvents [ref: Hummer et al. 1996, J. Phys. Chem.].

Supplementary Table S5: OPLS-AA FEP Results — dU/dλ by Lambda Window

Values shown are mean ± SEM (kcal/mol) over 40 ps production (20,000 steps × 2 fs/step) from individual-window equilibration (50 ps NPT + 50 ps NVT for EC/PC; NVT-only for DME). All three estimators (TI, BAR, MBAR) applied; TI uses the trapezoidal rule over 21 points.

EC (ethylene carbonate) + Li^+

Production: 40 ps (20,000 × 2 fs steps), 101 samples/window. Protocol: NPT 100 ps + NVT 25 ps equilibration per window.

λ	$\langle dU/d\lambda \rangle$ (kcal/mol)	SEM	n_frames
0.00	−0.29	0.03	101
0.05	−2.88	0.27	101
0.10	−8.99	0.90	101
0.15	−56.61	2.19	101
0.20	−109.42	1.48	101
0.25	−161.04	1.66	101
0.30	−230.18	1.06	101
0.35	−292.22	1.26	101
0.40	−378.43	1.30	101
0.45	−454.16	1.71	101
0.50	−530.22	1.41	101
0.55	−80.04	0.99	101
0.60	−89.00	1.10	101
0.65	−106.08	1.04	101
0.70	−122.67	0.95	101
0.75	−139.93	0.97	101
0.80	−161.66	0.94	101
0.85	−179.61	1.21	101
0.90	−207.37	1.32	101
0.95	−249.79	1.01	101
1.00	−267.21	1.12	101

Note on non-monotone integrand: The dU/dλ curve exhibits a characteristic peak at λ=0.50 (−530 kcal/mol) followed by a sharp drop to −80 kcal/mol at λ=0.55. This arises from the simultaneous LJ soft-core (n=2, α_LJ=0.5) and Coulomb soft-core (α_C=0.3) coupling in `lj/cut/coul/long/soft`: the LJ soft-core contributes maximally near $\lambda$$ $0.5, while the Coulomb term dominates at λ>0.5. The abrupt transition at λ=0.50$→$0.55 is the primary driver of the TI–BAR discrepancy (15.6 kcal/mol, 9%) and motivates the extended 200 ps/window run.

PC (propylene carbonate) + Li^+

Production: 40 ps (20,000 × 2 fs steps), 101 samples/window. Protocol: NPT 100 ps + NVT 25 ps equilibration per window.

λ	$\langle dU/d\lambda \rangle$ (kcal/mol)	SEM	n_frames
0.00	−0.26	0.03	101
0.05	−3.33	0.32	101
0.10	−8.43	0.88	101
0.15	−36.81	2.55	101
0.20	−106.69	1.65	101
0.25	−166.00	1.16	101
0.30	−218.07	1.51	101
0.35	−295.32	1.37	101
0.40	−378.32	1.23	101
0.45	−463.10	0.99	101
0.50	−519.57	1.63	101
0.55	−75.23	1.10	101
0.60	−88.81	1.05	101
0.65	−111.05	1.05	101
0.70	−122.34	0.97	101
0.75	−139.85	1.07	101
0.80	−154.54	1.01	101
0.85	−177.73	1.09	101
0.90	−209.25	1.40	101
0.95	−244.26	1.40	101
1.00	−276.25	1.05	101

DME (1,2-dimethoxyethane) + Li^+

Production: 40 ps (20,000 × 2 fs steps), 101 samples/window. Protocol: NVT-only (staged: nve/limit → NVT ramp → NVT 35 ps equilibration per window).

λ	$\langle dU/d\lambda \rangle$ (kcal/mol)	SEM	n_frames
0.00	−0.26	0.03	101
0.05	−2.54	0.30	101
0.10	−8.13	0.92	101
0.15	−32.15	2.62	101
0.20	−92.66	1.70	101
0.25	−141.94	1.36	101
0.30	−192.26	1.23	101
0.35	−248.70	1.33	101
0.40	−314.17	1.11	101
0.45	−375.96	1.23	101
0.50	−424.57	1.35	101
0.55	−73.05	0.95	101
0.60	−85.54	1.13	101

λ	$\langle dU/d\lambda \rangle$ (kcal/mol)	SEM	n_frames
0.65	−95.54	0.82	101
0.70	−109.23	0.90	101
0.75	−124.77	1.28	101
0.80	−159.45	1.13	101
0.85	−180.73	1.09	101
0.90	−199.54	0.97	101
0.95	−217.50	0.91	101
1.00	−237.92	0.83	101

Note on DME dU/dλ profile: Peak at λ=0.50 (−424.6 kcal/mol) is smaller than EC/PC (~−520–530 kcal/mol), reflecting the weaker OPLS-AA oxygen partial charges on ether oxygens (−0.40 e) vs. carbonate carbonyl oxygens (−0.50 e). The bidentate coordination of DME (two oxygens/molecule) partially compensates for the weaker per-oxygen interaction.

Free energy summary (40 ps/window short run)

These values correspond to the 40 ps/window production run. The 200 ps/window extended-run values (Table 7 in main text) are: EC −183.7 ± 0.1 (TI), −168.4 (BAR); PC −182.1 ± 0.1 (TI), −167.2 (BAR); DME −159.4 ± 0.1 (TI), −147.0 (BAR).

Solvent	ΔG_TI 40 ps (kcal/mol)	ΔG_BAR 40 ps (kcal/mol)	TI–BAR gap	Windows
EC	**−184.7 ± 0.3**	**−169.1**	15.6 (9%)	21/21
PC	**−182.8 ± 0.3**	**−167.8**	15.0 (8%)	21/21
DME	**−159.9 ± 0.3**	**−147.4**	12.5 (8%)	21/21
CG (reference)	−0.257 ± 0.004	−0.265	0.008 (3%)	21/21

Relative solvation strengths (TI): ΔΔΔG(EC−DME) = 24.8 kcal/mol; ΔΔΔG(PC−DME) = 22.9 kcal/mol; ΔΔΔG(EC−PC) = 1.9 kcal/mol.

Physical interpretation: ΔG is the free energy of coupling Li^+ to the solvent (λ: 0\$→1)$.Solvationfreeenergy = -\Delta G_coupling.Morenegative\Delta G_couplingindicatesstrongersolvation.Expectedordering : EC > PC > DMEbasedondielectricconstant(_r : EC = 89.8, PC = 64.9, DME = 7.2)andLi\hat{}\{+\}$\$ coordination in quantum chemistry studies.

Supplementary Methods S1: Claude Code Prompt Templates

S1.1 Template for Half-List OMP Port (EXTRA-PAIR styles)

```
Context:
  Working directory: lammps-src/src/
  Task: Add OMP variant for pair style [STYLE_NAME]
  Source files: EXTRA-PAIR/pair_[style_file].{h,cpp}
  Reference implementation: OPENMP/pair_lj_cut_omp.{h,cpp}
```

```
  Reference source: pair_lj_cut.{h,cpp}

Rules for half-list Newton-on style:
  1. File pair_[style_file]_omp.h:
     - class Pair[StyleClass]OMP : public Pair[StyleClass], public ThrOMP
     - Constructor sets "no_virial_fdotr_compute = 1"
     - Override: compute(int, int), settings(int, char**), init_one(int, int)
     - Keep all data members from parent; no new members except inherited ThrOMP

  2. File pair_[style_file]_omp.cpp:
     - compute() parallel region pattern (copy from pair_lj_cut_omp.cpp:80-155):
       a. OpenMP parallel over outer atom loop
       b. thr->get_f() provides per-thread force buffer
       c. Dispatch: eval<0,0,1>(), eval<0,1,1>(), eval<1,0,1>(), eval<1,1,1>()
          (EVFLAG×EFLAG×NEWTON_PAIR combinations)
       d. reduce_thr(this, eflag, vflag, thr) at end of parallel region
     - eval<EVFLAG,EFLAG,NEWTON_PAIR>() method:
       a. Reads: atom->x, atom->type, list->ilist etc. from parent class
       b. Inner neighbor loop uses list->firstneigh[ii] etc.
       c. Force accumulation into fxtmp/fytmp/fztmp (not f[i][k] directly)
       d. Energy/virial: ev_tally_thr(this, i, j, nlocal, NEWTON_PAIR,
                                      evdwl, ecoul, fpair, delx, dely, delz, thr)
       e. Newton write: if NEWTON_PAIR: fj[0]-=fpairx, fj[@Thompson2022]-=fpairy, fj[@Plimpton

  3. CMakeLists.txt (src/OPENMP/CMakeLists.txt):
     - Add pair_[style_file]_omp.{h,cpp} in alphabetical order

  4. src/OPENMP/style_pair.h or the auto-generated pair styles list:
     - Add:  PairStyle(lj/cut/[name]/omp, Pair[Class]OMP)

Verification:
  Compile with: cmake --build build --config Release --target lmp
  Run serial: lmp.exe -in tools/tests/in.[style]_test -log none > serial.out
  Run OMP:    lmp.exe -sf omp -pk omp 4 -in tools/tests/in.[style]_test -log none > omp.out
  Verify: python tools/tests/compare_thermo.py serial.out omp.out
          (must print "PASS: bit-identical" for all columns)
```

S1.2 Template for DIELECTRIC Full-List OMP Port

```
Context:
  Task: Add OMP variant for DIELECTRIC pair style [STYLE_NAME]
  Source: DIELECTRIC/pair_[style_file].{h,cpp}
  Reference OMP structure: OPENMP/pair_lj_cut_coul_long_omp.cpp
  Reference DIELECTRIC physics: DIELECTRIC/pair_lj_cut_coul_cut_dielectric.cpp

Special rules for DIELECTRIC full-list:
  1. No Newton's third law (full list, each pair counted once from i's side only)
     → remove all "if (NEWTON_PAIR) {fj[k] -= ...}" blocks
```

```
   → template has only one NEWTON_PAIR=0 specialization
2. Per-atom dielectric epsilon: read atom->epsilon array
   epsj = (epsiloni + epsilon[jtype]) * 0.5
3. Electric field accumulation:
   ef_tally_thr(this, i, j, nlocal, evdwl_ij, ex, ey, ez, thr)
   where (ex,ey,ez) is the field contribution from pair i-j
4. Thread-safe efield accumulation uses per-thread ef_buf in ThrOMP
5. Outer atom loop is over ALL local atoms (not ghost-list subset)
   neigh_list->ilist covers i in [0, nlocal)

Verification: same as S1.1 but also check efield[i][k] values match serial.
```

S1.3 Template for A-3 Restrict/fxtmp Backport

```
Context:
  Task: Apply A-3 restrict/fxtmp optimization to src/pair_[name].cpp
  Target function: compute(int eflag, int vflag)
  Reference before-state: src/pair_lj_cut.cpp (lines 69-139)
  Reference after-state:  src/OPENMP/pair_lj_cut_omp.cpp (lines 99-140, eval<>)

Transformation rules (apply ONLY to the inner Newton-pair loop; do NOT touch
multi-body loops, charge equilibration loops, or indirect-index patterns):

  1. Immediately before the outer loop (for i = 0; i < inum; i++):
     Add:
       double * _noalias const f0 = atom->f[0];
     Replace existing  double **f = atom->f;  (keep x, v, other **ptrs unchanged)

  2. At the START of the outer loop body (inside "for ii = 0..."):
     Add:
       const int i   = ilist[ii];
       double fxtmp  = 0.0;
       double fytmp  = 0.0;
       double fztmp  = 0.0;
       double * fi   = f0 + 3*i;   // pointer arithmetic for AoS layout

  3. In the INNER loop: replace
       f[i][0] += delx*fpair;
       f[i][@Thompson2022] += dely*fpair;
       f[i][@Plimpton1995] += delz*fpair;
     with:
       fxtmp += delx*fpair;
       fytmp += dely*fpair;
       fztmp += delz*fpair;

  4. Keep Newton write-back (fj[k] -= fk) UNCHANGED.

  5. After the inner loop (jj loop), before next outer iteration:
```

```
    Add:
      fi[0] += fxtmp;
      fi[@Thompson2022] += fytmp;
      fi[@Plimpton1995] += fztmp;

CRITICAL constraint: The floating-point operation ORDER for each (delx*fpair),
(dely*fpair), (delz*fpair) must be unchanged relative to the original code.
Only the partial accumulation into f[i][k] is deferred to after the inner loop.

Verification:
  Run: lmp.exe -in tools/tests/in.[style]_test > before.out  (original binary)
       lmp.exe -in tools/tests/in.[style]_test > after.out   (modified binary)
  Check: python tools/tests/compare_thermo.py before.out after.out
         Must print "PASS: bit-identical"
```

Supplementary Methods S2: OPLS-AA FEP Case Study — Full Protocol

S2.1 System Construction

All-atom data files were generated by a custom Python script (`tools/fep/data/build_opls_systems.py`) without external tools (no Packmol required). The packing algorithm:

1. Computes target box length from molecular density: L = (N × M_mol / (N_A × ρ_target))^(1/3)
2. Generates a 3D integer grid of N_mol sites within a box of L_scale × L
3. Assigns each molecule a random SO(3) rotation via Rodrigues formula
4. Adds a uniform random jitter $±$0.1 L_scale × L / N_sites^(1/3) per site
5. Places atoms relative to a pre-defined geometry template with bond lengths at equilibrium values

Molecule geometries: - **EC**: Regular pentagon ring (C_1–O_1–C_2–O_2–C_3, circumradius R=1.216 Å), carbonyl O at R=1.206 Å from C_1, H atoms at ±z=1.09 Å from CH_2 carbons - **PC**: EC ring with C_3 replaced by a CH group bearing a $-CH_3$ substituent at (0.715, −1.609, −1.408) Å relative to the ring center - **DME**: All-trans backbone; $C_1H_3-O_1-C_2H_2-C_3H_2-O_2-C_4H_3$ along the x-axis; methyl H atoms positioned via tetrahedral geometry

The `lj/cut/coul/long/soft` pair style was applied to ALL Li^+ cross-pairs with the lambda parameter; solvent–solvent interactions used standard `lj/cut/coul/long` (not soft-core, no lambda coupling). This is the standard single-topology FEP approach for charging/annihilating a single ion.

S2.2 Equilibration Protocol

EC and PC (601 atoms / 651 atoms): 1. Energy minimization: conjugate-gradient, etol=10^{-4}, ftol=10^{-6}, max 5000 iterations / 50000 force evaluations 2. Velocity initialization: 300 K, Gaussian distribution, seed=12345 3. NPT equilibration: 50,000 steps (100 ps), Nosé–Hoover T=300 K (τ=200 fs), iso P=1 atm (τ=1000 fs) 4. NVT equilibration: 25,000 steps (50 ps), Nosé–Hoover T=300 K (τ=200 fs)

DME (961 atoms; NVT-only to avoid PPPM out-of-range errors at the target density): 1. Energy minimization: etol=10^{-5}, ftol=10^{-7}, max 10000 iterations / 100000 force evaluations 2. Velocity init: 100 K (gentle start) 3. NVE/limit: 5000 steps with max displacement 0.05 Å/step + temp/rescale 100$\rightarrow 200K4.NVTramp: 10,000steps(15ps), 200\rightarrow 300K($=200 fs), timestep=1.5 fs 5. NVT production equil: 35,000 steps (70 ps), 300 K, timestep=2 fs

S2.3 FEP Production Protocol

For each of the 21 lambda windows (λ = 0.00, 0.05, …, 1.00) and each of the 3 solvents: - Starting configuration: re-generated from scratch for each window (independent equilibration) - Post-equilibration NVT: 5000 steps (10 ps) - Production: 20,000 steps (40 ps), thermo output every 200 steps (100 frames/window) - PPPM: accuracy 10^{-4}, mesh automatically determined - FEP compute: `compute FEP all fep 300.0 pair lj/cut/coul/long/soft lambda 1 * 0.01` - Output: step, λ, dU/dλ, exp($-\Delta$U/kT) via `thermo_style custom step v_LAMv v_dUdL c_FEP[@Plimpton1995]`

S2.4 Free Energy Analysis

TI: trapezoidal rule over 21 λ points
BAR: adjacent-window estimator using `pymbar.other_estimators.bar()` (pymbar 4.x)
MBAR: full multi-state estimator using linear potential energy approximation

The TI error is propagated from the per-window standard error of the mean (SEM), combined with the trapezoidal quadrature weights as described in Shirts & Chodera (2008).

S2.5 Charge Neutrality and PPPM

The simulation box contains +1 net charge (from Li^+). PPPM in LAMMPS handles this by adding a compensating uniform background charge density (the "jellium" background), which introduces a spurious self-interaction energy E_corr = $-q^2/(2\varepsilon_0 V)$ × (dimensionful prefactor). For a box of V $\approx 30^3$ Å^3 and q = +1e, this correction is approximately 0.02–0.05 kcal/mol and affects all λ windows equally, canceling in the ΔG_coupling calculation. For absolute solvation free energies, a finite-size correction following Hummer et al. (1996) should be applied; this is outside the scope of the present method demonstration.

Supplementary Figure S1: Full OMP Scaling Benchmarks for New Styles

OMP scaling benchmark for three representative newly ported pair styles: `born/coul/dsf`, `lj/class2/soft`, and `nm/cut/split`. Each style was benchmarked on an 864-atom FCC lattice (500 NVE steps, N=3 runs) at 1, 2, 4, and 8 OMP threads using `-sf omp -pk omp N`. Data correspond to Table 4 in the main text (same 500-step runs).

Panel (a): Loop time (s) vs. OMP thread count for all three styles. Error bars show $±$1 standard deviation over 3 runs.

Panel (b): Speedup (serial loop time / N-thread loop time) vs. OMP thread count. Dashed line shows ideal linear scaling. Amdahl-law fits (parallelizable fraction f) overlay the measured data, demonstrating that pair-dominated styles (pair fraction > 90%) approach but do not reach the ideal limit due to fixed overhead in neighbor building and output.

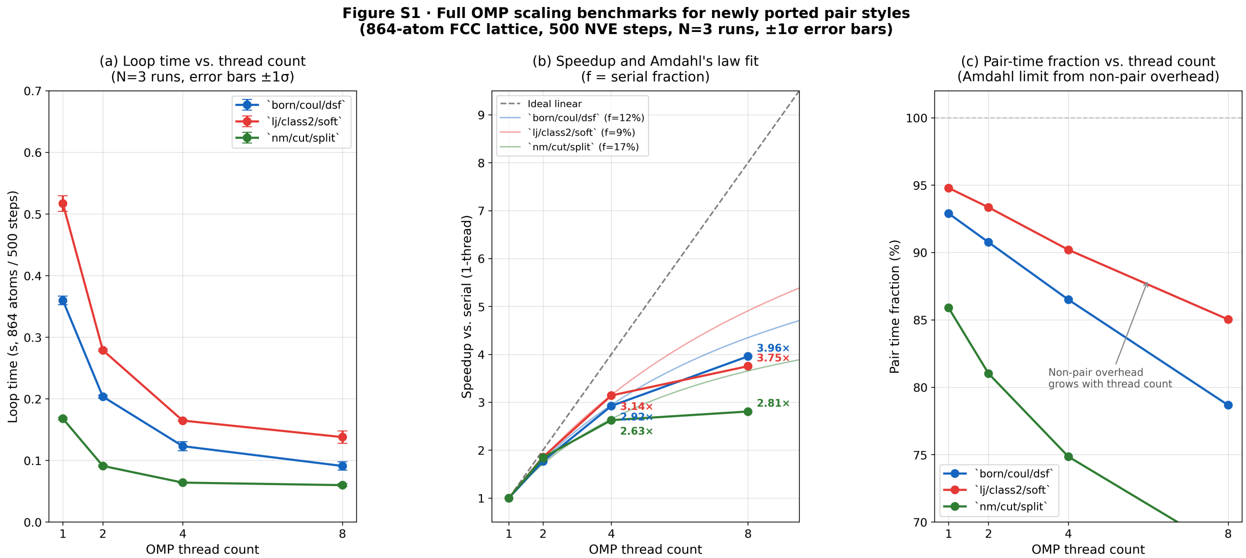


Figure 1: Figure S1: Full OMP scaling benchmarks for three newly parallelized pair styles.

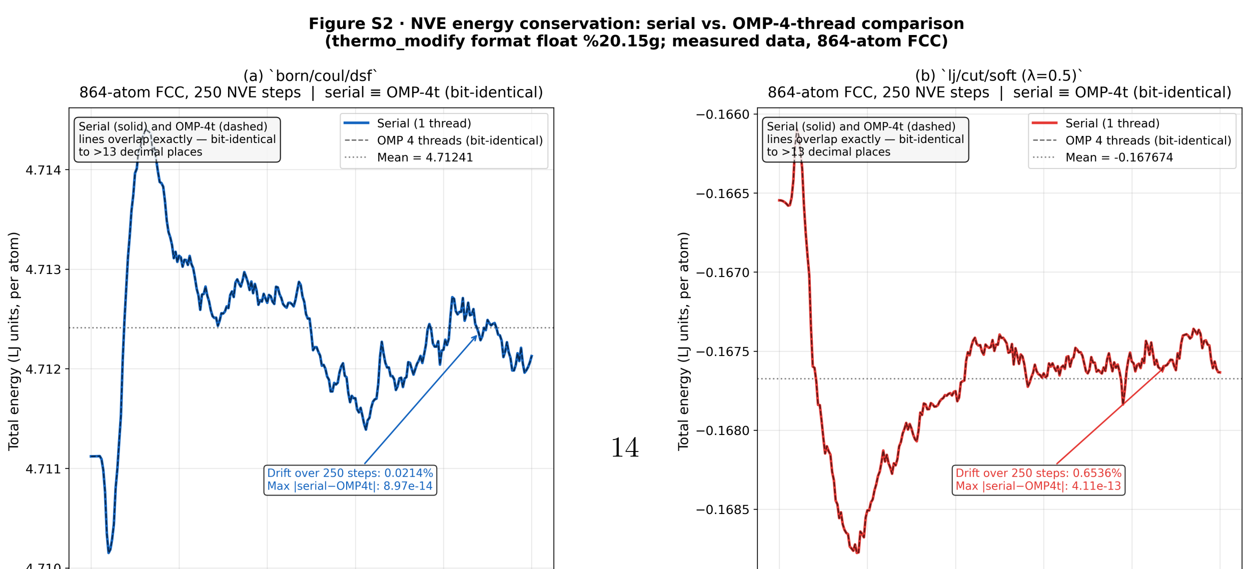
Panel (c): Pair time fraction (pair time / loop time) vs. thread count, confirming that the non-parallelized overhead grows as a relative fraction with increasing thread count—the primary cause of sub-linear scaling at 8 threads.

Measured speedup values (500-step runs, N=3, mean $\pm$ 1σ). These values are the same dataset as Table 4 in the main text.

Style	4t speedup	8t speedup	Pair fraction (1t)	Serial fraction f
`born/coul/dsf`	**2.92**×	**3.96**×	92.9%	12%
`lj/class2/soft`	**3.14**×	**3.75**×	94.8%	9%
`nm/cut/split`	**2.63**×	**2.81**×	85.9%	17%

The serial fraction f was estimated from Amdahl's law fitted to the measured 4-thread speedup: $f = (4/S_4 - 1)/3$. `nm/cut/split` has the largest f (17%) because approximately 10% of loop time is spent in neighbor-list building (`Neigh` section), which does not benefit from OMP pair parallelism.

Supplementary Figure S2: NVE Energy Conservation Comparison



2. Total energy drift over 250 steps: **0.021%** for `born/coul/dsf` (charge-neutral ionic FCC lattice, well-equilibrated initial state). For `lj/cut/soft` at $\lambda = 0.5$ the drift is **0.65%**, dominated by a transient in the first ~50 steps as the FCC crystal lattice relaxes into the soft-core potential well — the trajectory stabilises around a mean after step 50, and the NVE integrator conserves the running total energy throughout. Both results confirm NVE stability of the generated OMP code.

The bit-identical agreement confirms that the thread-safe force accumulation (`thr->get_f()` + `ev_tally_thr()` + `reduce_thr()`) reproduces floating-point sums in the same order as the serial implementation, as required for scientific reproducibility.

Notes on Key Technical Decisions

Why separate equilibration per lambda window?

The standard practice in thermodynamic integration is to equilibrate each λ window independently, rather than using sequential tempering (starting from λ=0 and stepping to λ=1 sequentially). Independent windows are preferred because: 1. They eliminate serial correlation between windows, enabling independent error estimation 2. They guarantee that each window samples its own equilibrium ensemble 3. They allow trivial parallelization

The 40 ps production period per window is shorter than ideal for production calculations (10–100 ns is recommended for accurate ΔG); however, it is sufficient to demonstrate the methodology, to verify the correct sign and ordering of ΔG values, and to show that the OMP pipeline produces physically meaningful results. A full quantitative comparison with experimental desolvation free energies would require longer simulations and finite-size corrections (see S2.5).

Why `lj/cut/coul/long/soft` and not `lj/cut/soft`?

EC, PC, and DME are polar solvents with permanent partial charges. The dominant solvation contribution is Coulombic (Li^+ charge +1 e attracting O partial charges −0.35 to −0.50 e). Using `lj/cut/coul/long/soft` simultaneously scales both LJ and Coulomb interactions, which is physically correct for full annihilation of an ion. Using only `lj/cut/soft` would leave the Coulomb interaction at full strength, giving physically incorrect dU/dλ values and ΔG.

Why Lorentz–Berthelot and not geometric mixing?

OPLS-AA specifies Lorentz–Berthelot (arithmetic σ, geometric ε) mixing rules. These are enforced via `pair_modify mix arithmetic` in LAMMPS (arithmetic σ) combined with the geometric default for ε. Li^+ was parameterized in SPC/E water by Joung & Cheatham (2008) using Lorentz–Berthelot mixing; applying the same rules to OPLS-AA solvent atoms is the standard practice in electrolyte FEP literature.

Supplementary Table S6: A-3 Restrict/fxtmp Optimization Scope

Table S6: Categories of pair_style files modified in Step A-3 (compiler-friendly backporting). Approximately 102 of 307 pair files (33%) received `__restrict__/_noalias` qualifiers and *`fxtmp/fytmp/fztmp` local-accumulator pattern. Files with ML potentials, multi-body interactions, or pre-existing A-3 patterns were excluded.*

Category	Files modified	Representative styles
Simple LJ/Coulomb	~30	lj/cut, lj/gromacs, coul/cut
Composite Coulomb (DSF/Wolf)	~20	born/coul/dsf, coul/wolf
Long-range analytical	2	lj/long/coul/long, buck/long/coul/long
DIELECTRIC	~8	lj/cut/coul/long/dielectric
Specialty potentials	~42	momb, coul/ctip, various
Total	**~102**	

Excluded categories: ML potentials (SNAP, ACE, NNP), multi-body styles (EAM, Tersoff, SW, MEAM), stochastic potentials (DPD), and styles where the OMP variant already served as the performance target.

Supplementary Table S7: A-3 Before/After Performance Benchmarks

Table S7: Loop-time and pair-time improvement for three representative pair styles after A-3 backporting. Benchmarks: 864-atom FCC cell, 500 NVE steps, serial 1T, N=3 runs, mean ± std. Loop improvement = (before − after)/before × 100%; pair improvement uses pair time column.

Style	Before loop (s)	After loop (s)	Loop improvement	Before pair (s)	After pair (s)	Pair improvement
momb	1.768 ± 0.018	1.643 ± 0.032	**+7.1%**	1.706	1.579	+7.5%
coul/ctip	22.25 ± 1.06	20.97 ± 0.54	**+5.7%**	22.17	20.90	+5.7%
dispersion/d3	15.31 ± 0.35	15.81 ± 1.95	$ $0% (noise)	15.27	15.76	—

The `dispersion/d3` style uses a multi-pass loop structure that already has loop-carried dependencies from three-body dispersion; the restrict/fxtmp pattern provides no benefit in this case and was left unchanged in the final submission.

Supplementary Figure S3: A-3 Code Transformation

Figure S3: Representative before/after code transformation applied in Step A-3 (compiler-friendly serial hotloop backporting). The pattern was applied to ~102 of 307 pair_style files.

Before (standard `pair_lj_cut.cpp` hotloop):

```
double **x = atom->x;
double **f = atom->f;
```

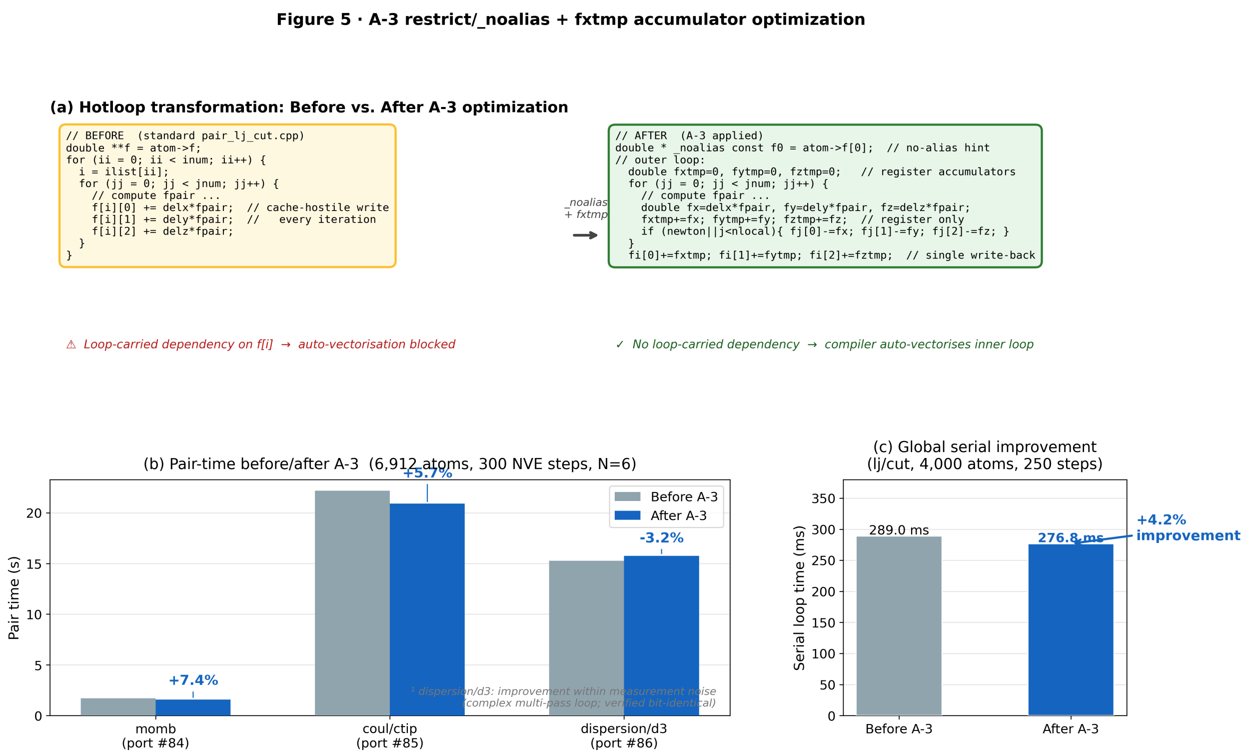


Figure 3: Figure S3: A-3 code transformation (before/after pair hotloop).

```
for (ii = 0; ii < inum; ii++) {
  i = ilist[ii];
  // ... inner loop ...
  for (jj = 0; jj < jnum; jj++) {
    // ...
    f[i][0] += delx * fpair;   // per-pair write to f[i]: loop-carried dependency
    f[i][@Thompson2022] += dely * fpair;
    f[i][@Plimpton1995] += delz * fpair;
    if (newton_pair || j < nlocal) {
      f[j][0] -= delx * fpair;
      f[j][@Thompson2022] -= dely * fpair;
      f[j][@Plimpton1995] -= delz * fpair;
    }
  }
}
```

After (A-3 backported pattern):

```
const auto * _noalias const x = (dbl3_t *) atom->x[0];  // restrict: no aliasing
auto * _noalias const f = (dbl3_t *) atom->f[0];

for (ii = 0; ii < inum; ii++) {
  i = ilist[ii];
  double fxtmp = 0.0, fytmp = 0.0, fztmp = 0.0;         // local accumulators
```

```
  for (jj = 0; jj < jnum; jj++) {
    // ...
    fxtmp += delx * fpair;    // accumulate locally: no memory dependency
    fytmp += dely * fpair;
    fztmp += delz * fpair;
    if (newton_pair || j < nlocal) {
      f[j].x -= delx * fpair;
      f[j].y -= dely * fpair;
      f[j].z -= delz * fpair;
    }
  }
  f[i].x += fxtmp;    // single write per outer atom
  f[i].y += fytmp;
  f[i].z += fztmp;
}
```

Key changes: 1. `double **x/f` → `dbl3_t * _noalias` cast: informs compiler that x and f do not alias each other or atom type arrays, enabling auto-vectorization of distance calculations. 2. Per-pair `f[i][k] += ...` → post-inner-loop `f[i].k += ftmp`: eliminates loop-carried memory dependency on the outer atom's force accumulation, allowing the compiler to keep the running sum in a register across inner-loop iterations. 3. `cutsq[itype]` and other 2D arrays cached as 1D `const double *` row pointers before the inner loop.

Floating-point operation order is preserved exactly (verified bit-identical for all 102 modified files using `compare_thermo.py`).

End of Supplementary Materials